

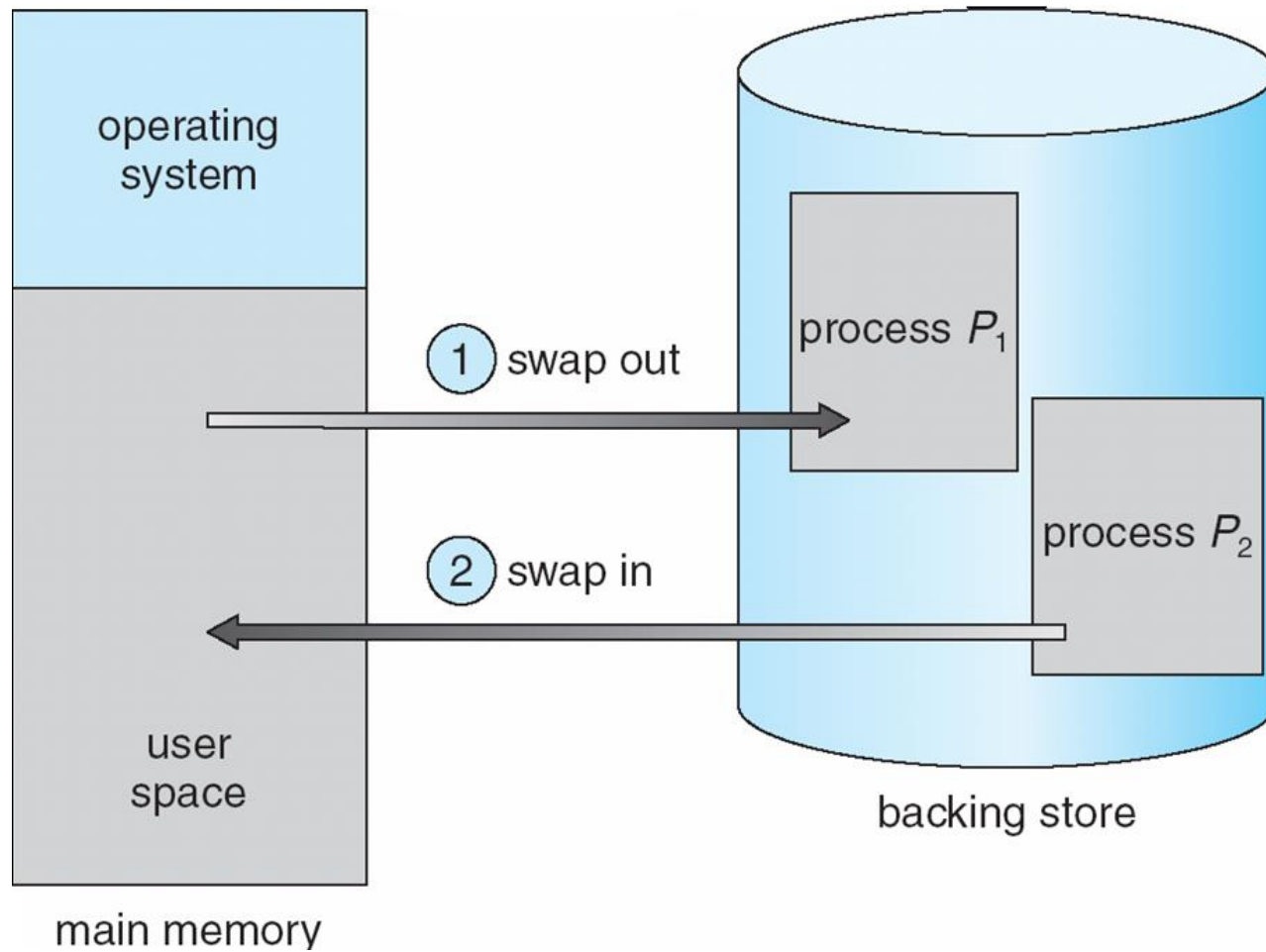
OPERATING SYSTEMS

Memory Management

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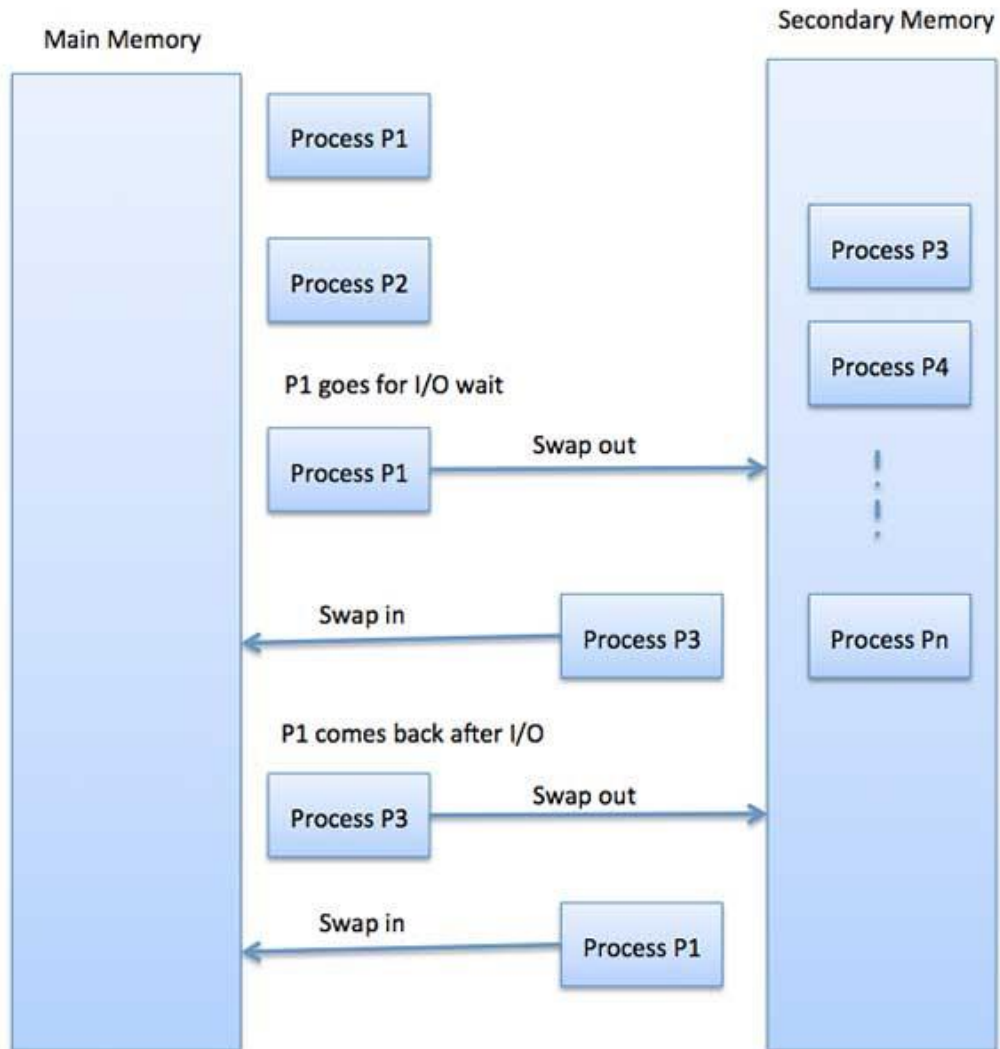
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- A process can be swapped temporarily out of memory to a backing store, and then brought back into memory for continued execution
- **Backing store** – fast disk large enough to accommodate copies of all memory images for all users; must provide direct access to these memory images
- **Roll out, roll in** – swapping variant used for priority-based scheduling algorithms; lower-priority process is swapped out so higher-priority process can be loaded and executed

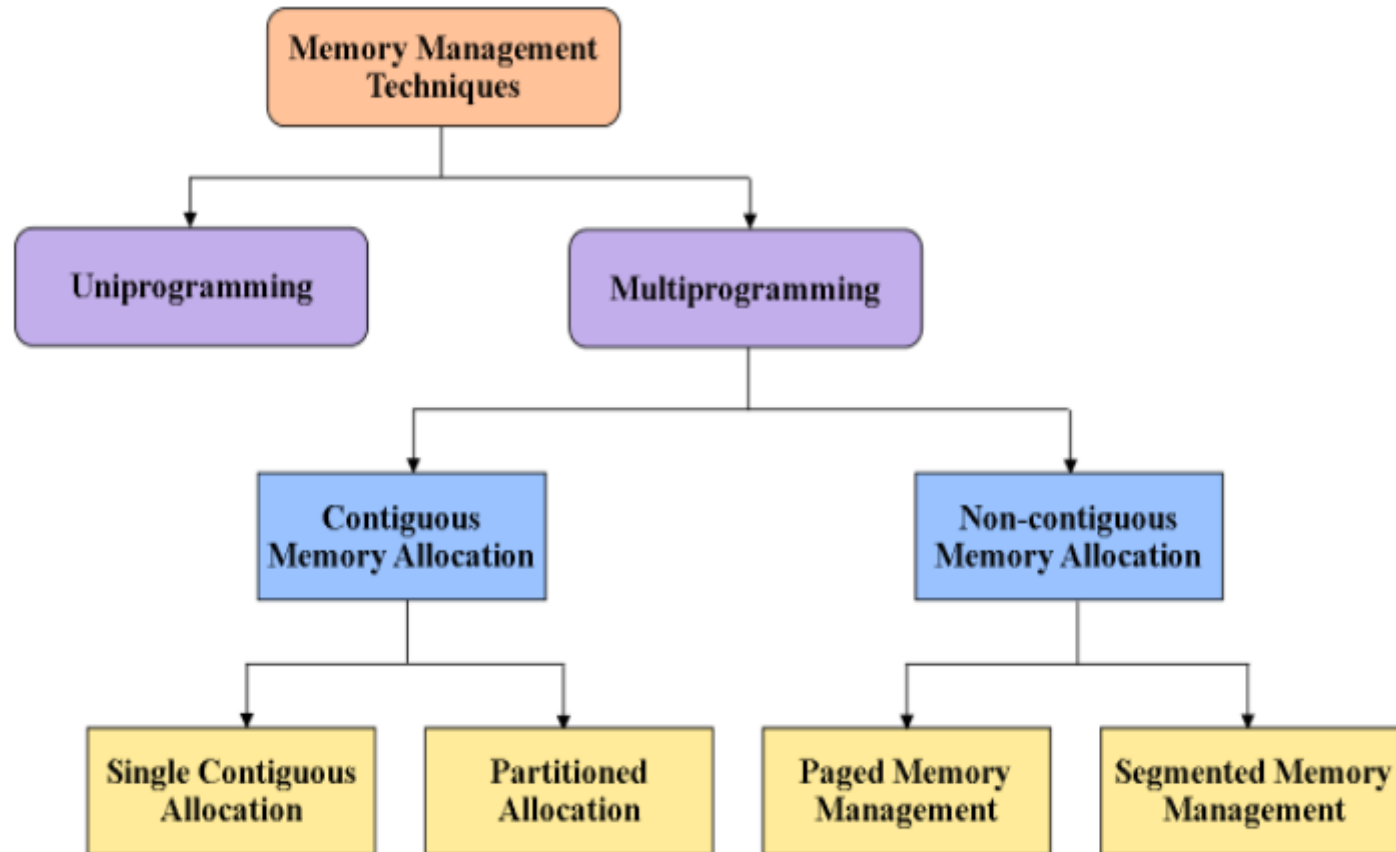


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Swapping



- **Swapping is a slow process** when compared to other operations
- Swapping increases the overhead
- Hence it should be reduced as much as possible
- Swapping should be done only when the processes are idle or don't have any pending i/o operations
- Use only OS buffers during i/o operations, instead of swapping
- Modern Oses (used in mobile phones) don't use swapping. It prompts the user to free up the memory space

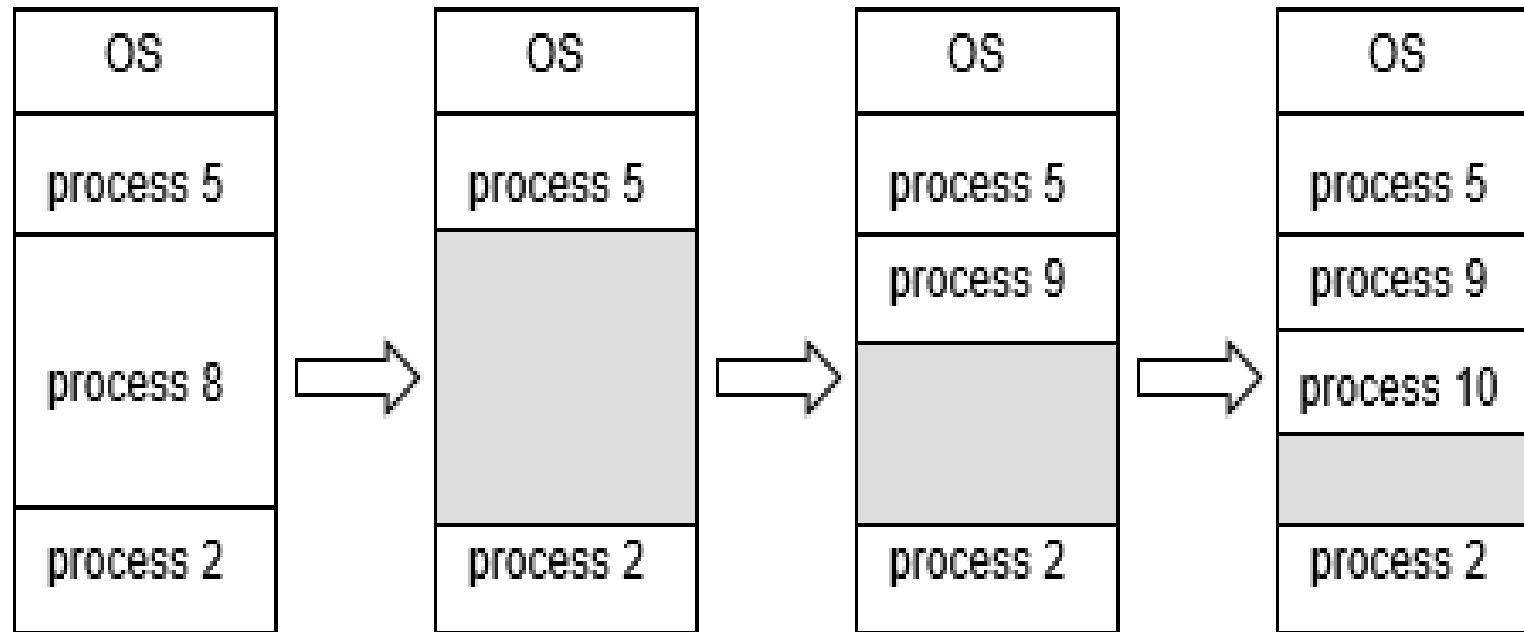


- Main memory usually divided into two partitions:
 - Resident operating system, usually held in low memory with interrupt vector
 - User processes then held in high memory
- Single partitioned contiguous memory allocation
 - Entire memory is available for a single application
 - A small portion is left for the OS and rest is used for user applications
 - Example: MS-DOS
 - Not much used in present day OSes

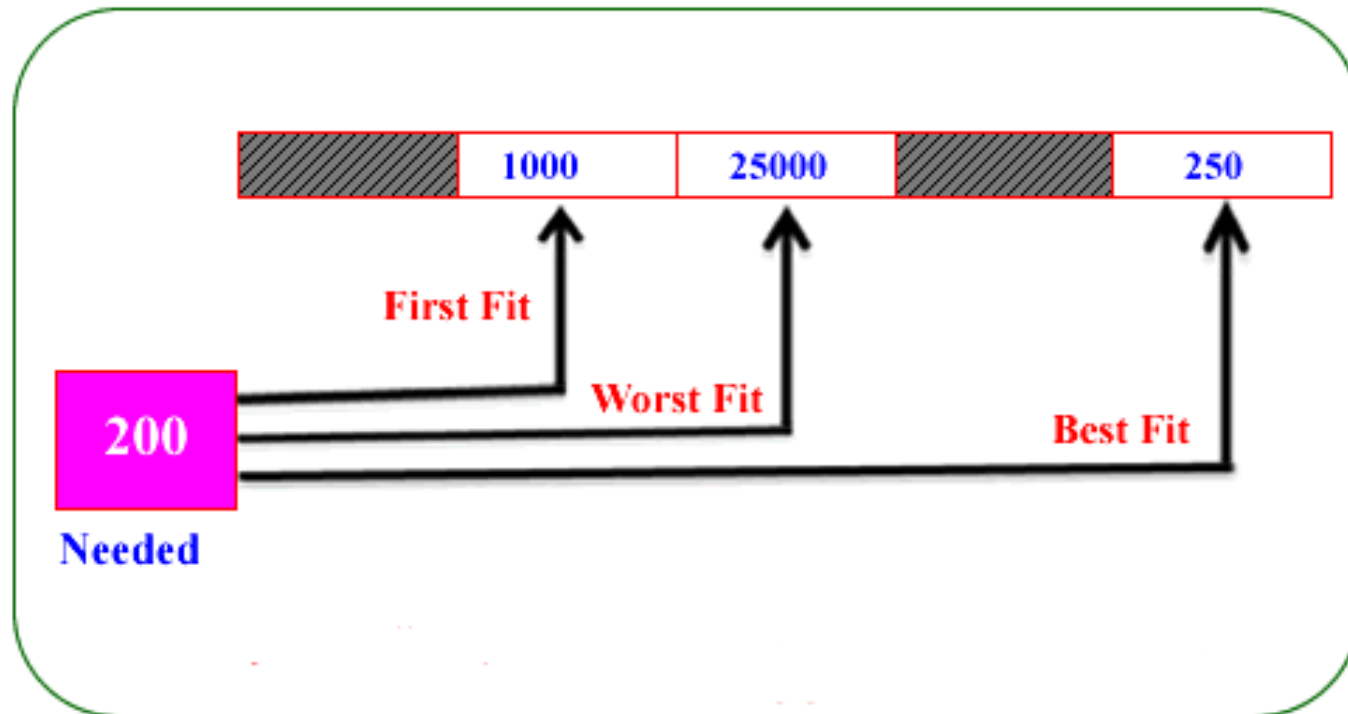
- Multiple partitioned contiguous memory allocation
 - Divide all available memory into multiple partitions
 - Assign each process to a partition
 - This can satisfy the requirement of multiple programs accommodated in memory
 - OS keep a information about the allocated partitions and free partitions
 - **Unused memory block is called as holes**

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Contiguous Memory Allocation



- When a process arrives, it is allocated memory from a hole large enough to accommodate it
- OS may use different strategies to find a hole to fit the process
 - **First-fit**: Allocate the *first* hole that is big enough
 - **Best-fit**: Allocate the *smallest* hole that is big enough; must search entire list, unless ordered by size
 - **Worst-fit**: Allocate the *largest* hole; must also search entire list
- First-fit and best-fit better than worst-fit in terms of speed and storage utilization

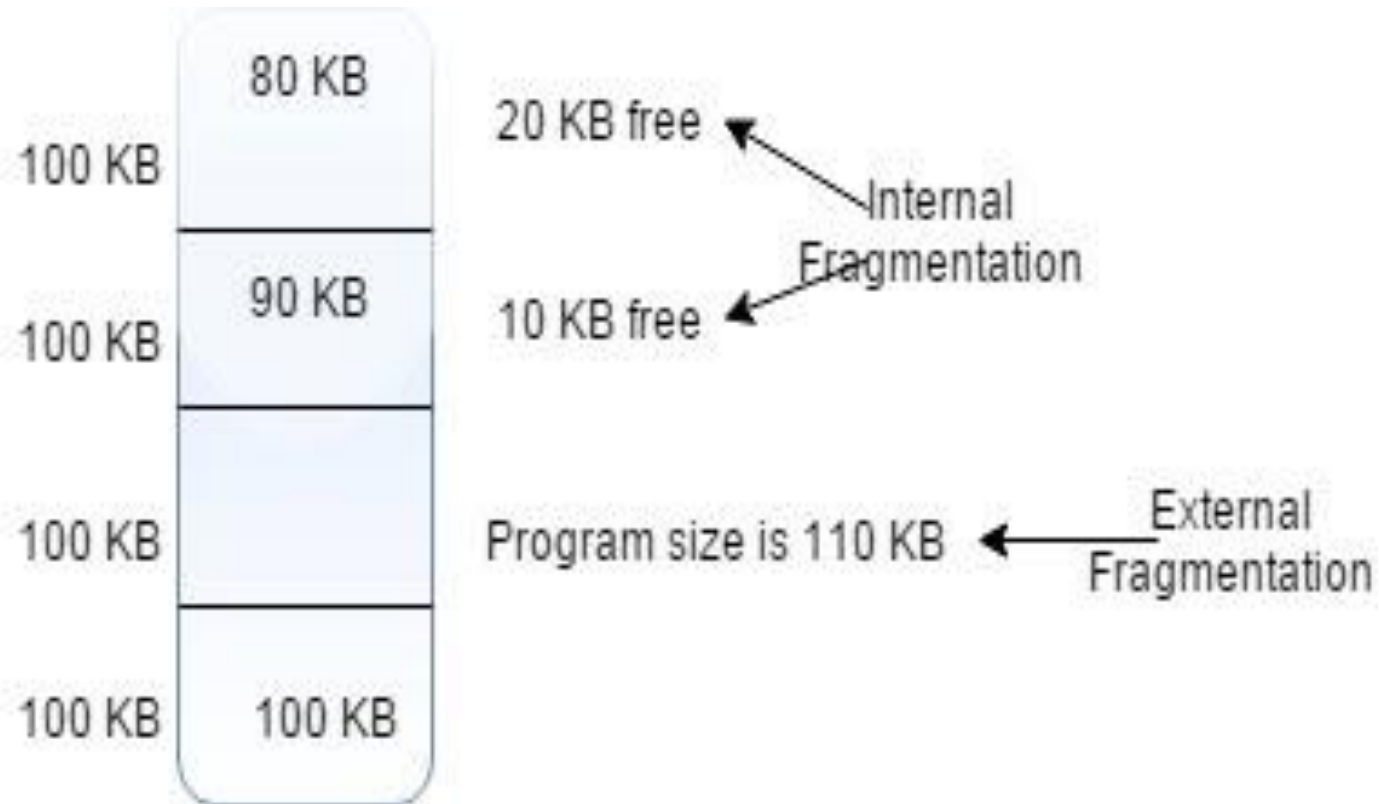


Fragmentation

- **External Fragmentation** – total memory space exists to satisfy a request, but it is not contiguous
- **Internal Fragmentation** – allocated memory may be slightly larger than requested memory; this size difference is memory internal to a partition, but not being used
- Reduce external fragmentation by **compaction**
 - Shuffle memory contents to place all free memory together in one large block
 - Compaction is possible *only* if relocation is dynamic, and is done at execution time
 - May lead to I/O problem

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Contiguous Memory Allocation



Suppose a process requests 12KB of memory and the memory manager currently has a list of unallocated blocks of 6KB, 14KB, 19KB, 11KB, and 13KB blocks. Where does the 12KB process will get fitted using different allocation strategies?

6KB
14KB
19KB
11KB
13KB



THANK YOU

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